

Homework 5: Multiple Linear Regression

STAT 141, Week 5

Your name here

Due: Thursday, October 8th at 9:00am on Gradescope as a PDF

In this homework assignment you will:

- Practice interpreting regressions with categorical predictors
- Practice using visualizations to perform EDA and inform regression models
- Practice applying transformations to help satisfy LINE assumptions
- Build and interpret linear regression models with fixed and varying slopes

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Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

Run the chunk below to load the necessary packages.

```
library(tidyverse)
library(ggglm)
library(moderndiver)
library(pdxTrees)
```

Exercises

Exercise 1: Factors (Categorical Predictors)

We will again work with the `pdxTrees` data package. Run the following code chunk to obtain a data frame of the four parks closest to Reed, subsetted to only include trees of the **Genus** *Pseudotsuga*, *Acer*, *Quercus*, or *Cornus*. There are some additional modifications to clean up the data. The chunk also shows how many trees of each **Genus** are in the `near_reed` dataset.

```
near_reed <- get_pdxTrees_parks() %>%
  filter(Park %in% c("Crystal Springs Rhododendron Garden",
                    "Kenilworth Park",
                    "Eastmoreland Garden",
                    "Berkeley Park")) %>%
  mutate(Type = case_when(
    Functional_Type == "BD" | Functional_Type == "CD" ~ "Deciduous",
    Functional_Type == "ED" | Functional_Type == "CE" ~ "Evergreen"
  )) %>%
  mutate(Crown_Width = (Crown_Width_NS + Crown_Width_EW)/2) %>%
  drop_na(Type) %>%
  filter(Genus %in% c("Pseudotsuga", "Acer", "Quercus", "Cornus"))

near_reed %>% group_by(Genus) %>% summarize(Count=n())
```

```
# A tibble: 4 x 2
  Genus      Count
  <chr>    <int>
1 Acer         104
2 Cornus        36
3 Pseudotsuga  193
4 Quercus       52
```

a. In the chunk below, construct a linear model to analyze the differences in `Carbon_Sequestration_lb` by `Genus`. **How many indicator variables are displayed in the regression table, and what are they? Which level of the `Genus` variable does not have an indicator variable?** [Code and text for this question]

b. Write down the formula for the linear model for `Carbon_Sequestration_lb` as a function of `Genus`. [Only text/math for this question]

c. Using *data wrangling* functions, compute the mean `Carbon_Sequestration_lb` for trees in each level of `Genus`. Then, identify where the value for the `Acer` genus appears in the regression table. [Code and text for this question]

d. Demonstrate how the mean `Carbon_Sequestration_lb` for trees from the `Cornus`, `Pseudotsuga`, or `Quercus` genus can be calculated from the *regression table*. [Text/math for this question]

e. Given your findings from parts (b)-(d), carefully interpret the coefficient for the `Cornus` genus. [Only text for this question]

f. By default, R chooses the alphabetically first level of the explanatory variable to serve as the baseline. However, the order can be changed in R using some extra code. In the following code chunk, we re-order the `Genus` variable so that the largest category, `Pseudotsuga`, is the *baseline* group.

```
near_reed <- near_reed %>%  
  mutate(Genus = fct_relevel(Genus, "Pseudotsuga", "Acer", "Quercus", "Cornus"))
```

Create a new linear model for `Carbon_Sequestration_lb` as a function of `Genus`. Comment on how and why the coefficients estimates have changed. [Code and text for this question]

g. Suppose a particular tree is a Bigleaf Maple, which is in the *Acer* genus. **What is the predicted Carbon Sequestration using the model constructed in part (a)? What is the predicted Carbon Sequestration using the model constructed in part (f)? How do the predictions compare?** [Text/math only for this question]

Exercise 2: Regression with Transformation

Problems 2 & 3 use an extended version of the [Opportunity Insights dataset](#) (introduced in the activity from the *Data Wrangling* lecture) that has been joined with a dataset from The National Association of College and University Business Officers containing information on university endowment assets from 2016.¹ The data are in the file `colleges_endow.Rdata`; for simplicity, the data frame is named `colleges`. The following table provides a list of the variables in the dataset and their descriptions. (Note: It is easier to read the table in the Rendered PDF document.)

¹Specifically, the endowment data are from fiscal year 2016, which began on June 1, 2015 and ended on May 31, 2016. For the purposes of Stat 141, it is fine to simply refer to these data as being collected in 2016.

Variable	Description
name	name of college
region	region, 1 for Northeast, 2 for Midwest, 3 for South, and 4 for West
state	state
tier_name	tier defined by selectivity and type
type	coded 1 for public, 2 for private non-profit, 3 for for-profit
exp_instr_pc_2013	instructional expenditures per student in 2013
ipeds_enrollment_2013	total undergraduate enrollment in Fall 2013
sticker_price_2013	average annual cost of attendance in 2013
scorecard_netprice_2013	net annual cost of attendance for bottom income quintile in 2013
grad_rate_150_p_2013	percentage of students graduating within 150% of normal time in 2013
avgfacsal_2013	average faculty salary in 2013
sat_avg_2013	average SAT scores (scaled to 1600) in 2013
endowment_pc_2000	endowment assets per student in 2000
mr_kq5_pq1	mobility rate, top 20% of the income distribution
mr_ktop1_pq1	mobility rate, top 1% of the income distribution
par_median	median parent household income
par_q1	fraction of parents in first (bottom) income quintile
par_q2	fraction of parents in second income quintile
par_q3	fraction of parents in third income quintile
par_q4	fraction of parents in fourth income quintile
par_q5	fraction of parents in fifth (top) income quintile
par_top5pc	fraction of parents in top 5% of income distribution
par_top1pc	fraction of parents in top 1% of income distribution
k_median	median child individual earnings in 2014 (at age 34)
k_top5pc	fraction of children in top 5% of income distribution
k_top1pc	fraction of children in top 1% of income distribution
fy2016_endowment_pc	endowment assests per student in 2016
fy2016_endowment_thousands	total value of endowment in thousands USD in 2016

Run the following R chunk to load the `colleges` data frame.

```
load("data/colleges_endow.Rdata")
```

One metric of college access is net tuition price for low- and middle-income students after accounting for financial aid. The `scorecard_netprice_2013` variable contains data on the net annual cost of attendance for students with parental household income in the bottom quintile (in 2013); these data come from the U.S. Department of Education's College Scorecard.

A university's endowment refers to the financial assets invested by a university to support its educational and research mission; individuals and organizations can donate to contribute to a

university's endowment. The `fy2016_endowment_pc` variable contains data on the monetary value of endowment assets per student (in 2016), measured in USD.

Let's investigate the association between college access and endowment assets by building a model for college access using endowment assets.

a. We'll explore both the net tuition price variable and the endowment assets variable individually using **R**. What kind of graphs and summary statistics would be useful for doing this? [Text only for this question]

b. Use **R** to implement the graphs and summary statistics you discuss in part (a) to explore the distribution of each of these variables. **Describe the distribution of each variable in the context of the data, referencing both your numerical and graphical summaries.** [Text and code for this question]

c. Construct a plot showing the bivariate relationship between net tuition price and endowment assets. Be sure to map the response variable to the *y*-axis and the explanatory variable to the *x*-axis. [Code only for this question]

d. Based on the plot, **does there seem to be a relationship between college access and endowment assets? Does a line seem like a reasonable summary for the relationship between these variables?** Explain your reasoning. [Text only for this question]

e. Add a variable to the `colleges` dataset that contains the log-transformed version of `fy2016_endowment_pc` and name it `log_endow_pc`. That is, if `fy2016_endowment_pc` equals \$1000, then the log-transformed version equals $\log(1000) = \ln(1000) = 6.908$. Statisticians tend to refer to natural log as just “log”. The function in R to compute the (natural) log is `log()`. [Code only for this question]

f. Recreate the plot from part (c), but use the log-transformed version of `fy2016_endowment_pc` rather than the original variable. [Code only for this question]

g. Based on the plot in part (f), does a line seem like a reasonable summary of the relationship between net tuition price and log-transformed endowment assets? If so, calculate the correlation coefficient and interpret the sign and magnitude. [Code and text for this question]

h. Construct a simple linear regression model for net tuition price using the log-transformed endowment assets variable. **Print out the model coefficients table.** [Code only for this question]

i. Check the model diagnostics for the model from part (h), and **comment on them.** [Code and text for this question]

Exercise 3: Regression Models with Equal vs Varying Slopes

In this problem, we will continue working with the same data set from Exercise 2 to investigate whether the relationship between net tuition price for low- and middle-income students and endowment value seems to differ for Ivy Plus schools versus other schools.

a. Using *data wrangling* functions, create a new variable that categorizes whether or not a school is Ivy Plus. [Code only for this question]

b. Construct a plot showing the association between net tuition price and log-transformed endowment assets; color the points based on whether a school is in the Ivy Plus. (1) Describe what you see, then (2), using data wrangling functions, assess how the average net tuition price for these students at Ivy Plus schools compares to other schools.** [Code and text for this question]

c. Construct a multiple linear regression model for net tuition price using log-transformed endowment assets *and* whether a school is categorized as Ivy Plus. *Assume equal slopes in the model.* Print the model coefficients table. [Code only for this question]

d. Interpret each model coefficient from 3(c) in the context of the data. Be sure to comment on whether the intercept makes sense in this context. [Text only for this question]

e. In part (b), you used data wrangling functions to estimate the difference in average net tuition price for low- to middle-income students at schools in the Ivy Plus versus schools not in the Ivy Plus. **Explain the reasoning behind why this difference is not equal to the coefficient for the Ivy Plus predictor variable in the model from part (c).** [Text only for this question]

f. Construct a linear regression model for the net tuition price using log-transformed endowment assets and whether a school is categorized as Ivy Plus. *Assume varying slopes in this model.* [Code only for this question]

g. Based on the model from 3(f), **write out the formula for the regression line for just Ivy Plus schools and the formula for the regression line for just non-Ivy Plus schools.** Using these formulas, state **what is the estimated slope for schools in the Ivy Plus and what is the estimated slope for non-Ivy Plus schools.** [Text only for this question]

h. Produce a plot that illustrates the linear model fit in part (f) and analyzed in (g). Based on this plot and your analysis, is there evidence suggesting that making a donation to a school's endowment increases access for low- and middle-income students to the same extent at Ivy Plus schools as non-Ivy Plus schools? **Explain your reasoning.** [Code and text for this question]

(Hint: `geom_smooth()` may be useful here.)

Exercise 4: A Bigger Model for Big Trees

For this problem, we'll return to the `pdxTrees` dataset once more.

Suppose that we wish to predict tree height using multiple linear regression. Ideally, we want our model to be parsimonious, accurate, and sensible at the same time. This kind of task typically does not have one correct answer - there may be many sufficient models. The model created in the chunk below is one example of a model we could create for `tree_height`.

```
# Grab trees near Reed
near_reed <- get_pdxTrees_parks() %>%
  filter(Park %in% c("Crystal Springs Rhododendron Garden",
                    "Kenilworth Park",
                    "Eastmoreland Garden",
                    "Berkeley Park"),
         !is.na(Functional_Type)) %>%
  mutate(Crown_Width = (Crown_Width_NS + Crown_Width_EW)/2)

big_model <- lm(Tree_Height ~ Carbon_Sequestration_lb + DBH + Crown_Width +
                Native*DBH, near_reed)
get_regression_table(big_model)
```

```
# A tibble: 6 x 7
  term                estimate std_error statistic p_value lower_ci upper_ci
  <chr>              <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 intercept          7.14      1.93      3.70      0        3.34     10.9
2 Carbon_Sequestration_lb -0.087  0.049    -1.79    0.074   -0.183    0.009
3 DBH                 2.37     0.164    14.4      0        2.04     2.69
4 Crown_Width         0.071   0.081     0.881   0.379   -0.088    0.231
5 Native: Yes         9.44     3.07     3.08    0.002     3.41    15.5
6 DBH:NativeYes       0.153   0.137     1.12   0.264   -0.116    0.422
```

```
get_regression_summaries(big_model)
```

```
# A tibble: 1 x 9
  r_squared adj_r_squared  mse  rmse sigma statistic p_value  df  nobs
  <dbl>      <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl>
1  0.816      0.814  326.  18.0  18.1    559.      0     5  637
```

a. Write out the equation for this fitted model. [Only text/math for this question.]

b. Based on the information provided from the chunk at the beginning of the exercise, discuss whether you think this is a “good” model. Consider in particular our goals of having a model that is both parsimonious and accurate. [Text and code for this question]

c. If we were to compare this model to others with many more variables, should we use R^2 or adjusted R^2 ? [Text only for this question]

d. What is the difference in how we should interpret the coefficients on `DBH:NativeYes` versus `DBH`? [Text only for this question]
